

SHUBHAM BOLLU

CONTACT

- Vitry-sur-Seine, Paris, France
- +33 7 59 48 93 43
- shubhambollu1@gmail.com
- www.linkedin.com/in/shubham-bollu

Portfolio: https://portfolio-robotics-shubham.netlify.app/

GitHub: https://github.com/Shubhu1-ai

EDUCATION

École Supérieure d'Ingénieurs Léonard-de-Vinci (ESILV) - Paris, France
Masters Robotics & Industry
September 2025 - (expected 2027)

NK Orchid Engineering College - India
Bachelor's degree
Mechanical Engineering (2019-2022)
CGPA: 9.31/10

TECHNICAL SKILLS

Robotics & Control: 3D**M3DEXPERERINCE, MATLAB, SOLIDWORKS, Simulink, ROS, Arduino, Unity, VHDL, Robot simulation, inverse kinematics, trajectory planning**

AI & Machine Learning:
Python (NumPy, Scikit-learn, OpenCV), C, SQL, Machine-learning workflows

Data & Visualization: **Power BI, Tableau, SQL Server, SSMS, Azure, Linux.**

Industrial & Project Management: **Technical troubleshooting, 5S, Kaizen, Quality Control, Agile Methodology, Client communication & technical support**

LANGUAGES

- English (C1)
- French (A2)
- Hindi
- Telugu (Native)

Robotics & Industry 4.0 Master's student at Pôle Léonard de Vinci, Paris Paris with experience in industrial automation (Bosch), data center control systems (AWS) and data automation (TCS). Interested in robot kinematics, trajectory planning, system modelling and virtual validation of industrial systems. Two Dassault Systèmes–sponsored academic projects; building 3DEXPERIENCE skills. Seeking a 6-month end-of-studies internship in robotics simulation, Physical AI or digital twins.

WORK EXPERIENCE

Amazon Web Services (AWS), Paris
Data Center Operations Engineer Intern *April 2026 – August 2026 (4 months)*

- Built digital operational workflows and readiness dashboards using internal AWS data analytics tools, streamlining maintenance tracking and technical audits.
- Monitored **PLC logic** and **PID** feedback loops via Building Management Systems (**BMS**) and Electrical Power Monitoring Systems (**EPMS**), ensuring 99.999% uptime for Data Center infrastructure.
- Supported **vendor management**, risk mitigation and **KPI** reporting including **PUE** and MTTR across AWS **cloud infrastructure** serving European operations.

Tata Consultancy Services
Data & Automation Engineer *June 2023 - August 2025 (2 years and 2 months)*

- Designed and maintained **Power BI pipeline dashboard** to monitor dataflows, presented insights to international clients and leadership teams to secure and support dataflows.
- **Led** an **Oracle to SQL Server** migration for a large application, cutting database licensing costs by **35%**, and significantly improving performance.
- Supported Linux and Windows infrastructure and SQL Server applications for international oil and gas clients, investigating technical incidents and coordinating resolutions.

Bosch Chassis Systems India Pvt. Ltd
Machining & Assembly Supervisor *Jan 2023 - May 2023 (5 months)*

- Supervised assembly line operations for ABS wiring products and implemented **5S standards,** conducted daily quality inspections by coordinating with teams to ensure timely delivery.
- Implemented **Kaizen** that increased production efficiency by **10%** and reduced defects.

ACADEMIC & TECHNICAL PROJECTS

- **UR5 Robot Dynamics & Control:** Modelled and Simulated 6-DOF Universal Robots using MATLAB & Simulink. Implemented **inverse kinematics** and **trajectory planning, tested** in FABLAB.
- **ALLEGRI (prize winner; Hôpital Delafontaine de Saint-Denis partnership):** Built a YOLOv11 detection system running on a Raspberry Pi 5 for real-time inference, with a web and Android app and a override step. Reached 92.9% mAP50 for bacteria and 87.9% for malaria detection.
- **Robot work-cell simulation in 3DEXPERIENCE (in progress):** layout comparison, cycle time and reach analysis.

RESEARCH & INNOVATION PROJECTS

- **Solar Milk Pasteurization (Dassault Systèmes–sponsored academic project):** Developed solar-powered pasteurization process with temperature control and packing system for dairy processing. Used SOLIDWORKS and managed to maintain up to –4-degree temperature drops.
- **Solar DC Water Pump (Dassault Systèmes-Sponsored academic Project):** Designed a standalone solar-powered DC water pumping system for agricultural applications, covering component sizing, system design, and pumping-performance considerations.
- **Solar Mud Pot Air Cooler (Published Research Paper):** Designed solar cooler with natural air flow cooling mud pot and ran CFD simulation of thermal performance, temperature and airflow.

CERTIFICATIONS

- IBM watsonx.ai Data Science & ML Essentials (2025) - IBM
- Learning Industrial Automation - LinkedIn Learning (2025)
- Online Course on MATLAB and Simulink - NIELIT Calicut, India (2025)
- Data Visualization: Storytelling - LinkedIn Learning (2025)